

# Significant Error: Citations to Research With Publicized Statistical Errors

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## Abstract

Publicizing a statistical error could sharply reduce reliance on affected papers or prompt researchers to qualify their citations. We examine both responses after a 2011 critique of a common mistake in neuroscience: inferring that two effects differ because one is statistically significant and the other is not. Papers flagged for the mistake continued to receive citations well beyond a handful of highly cited cases. Their median annual citations rose from 5 in 2010 to 13–17 during 2012–2015; comparison papers rose from 4 to 10–11. In a separate sample of citations to flagged papers, 94 of 95 valid completed ratings recorded no acknowledgment of concerns. These findings document widespread continued use with little recorded qualification. Comparisons with unflagged papers leave the size of any citation penalty uncertain; continued citation does not establish that publicity had no effect. We extend the comparison to audits of 67 political-science papers and 200 animal studies. In the animal-study audit, mean annual citations changed little in both groups, with no clear additional decline among papers flagged for pseudoreplication. The additional audits broaden the evidence of continued use while leaving the size of any publicity effect uncertain.

We thank Danielle Portnoy and Xiaoran Huang for research assistance, and Andrew Gelman, Kabir Khanna, and Daniel Stone for comments. Replication materials: [https://github.com/recite/sig\\_fault](https://github.com/recite/sig_fault).

# 1 Introduction

Publicizing a statistical error gives researchers a reason to reconsider their reliance on affected findings. A substantial response could take two forms: fewer citations to papers containing the error, or more citations that acknowledge the problem. We examine both after a prominent methodological critique in neuroscience.

The mistake is to treat a statistically significant effect in one group and a nonsignificant effect in another as evidence that the groups differ. That conclusion requires a test of the difference itself. Significance in one group can reflect greater precision rather than a larger effect (Gelman and Stern, 2006). The missing test leaves the claimed contrast unsupported by that argument; it does not establish that every conclusion in the paper is false.

Nieuwenhuis et al. (2011) identified 157 papers in which a comparison of effects was relevant and classified 79 as making the mistake. They supplied us with the paper-level classifications. Researchers who understand the critique can recognize the mistake in an original paper and reconsider its use. The critique therefore need not name each affected paper to influence citations.

The affected papers remained widely cited, and recorded qualifications were rare. Median as well as mean citations increased after the critique; 69 of 76 flagged papers received more citations per year in 2012–2015 than in 2010. Almost all valid completed ratings in our citation-context sample recorded no concern. The comparison with unflagged papers is less conclusive about how much publicity changed citation growth. We distinguish that causal question from the observed persistence of largely unqualified citation.

# 2 Data and comparison

The source review covers behavioral, systems, and cognitive neuroscience papers published in 2009–2010 in *Nature*, *Science*, *Nature Neuroscience*, *Neuron*, and sampled issues of the *Journal of Neuroscience*. The classification concerns the particular statistical inference reviewed, not the validity of every result in a paper.

The archived citation exports cover 155 of the 157 classified papers. Two further histories contain the intended cited paper itself and records predating it, indicating invalid citation searches. Excluding these histories leaves 76 flagged and 77 comparison papers. We count distinct citing documents per paper and year, retain citation-free years within covered histories, and exclude the incomplete 2016 export. Appendix A documents coverage, duplicates, and linkage checks.

The critique appeared online on August 26, 2011. We use 2010 as the baseline and 2012–2015 as the post-publicity period, omitting the transition year 2011. The main regression is Poisson pseudo-maximum likelihood with article and calendar-year fixed effects. It compares proportional citation changes in flagged and comparison papers; uncertainty is clustered by article. A relative post/pre citation ratio below one indicates slower proportional growth for flagged papers. Appendix B gives the specification and alternative comparisons.

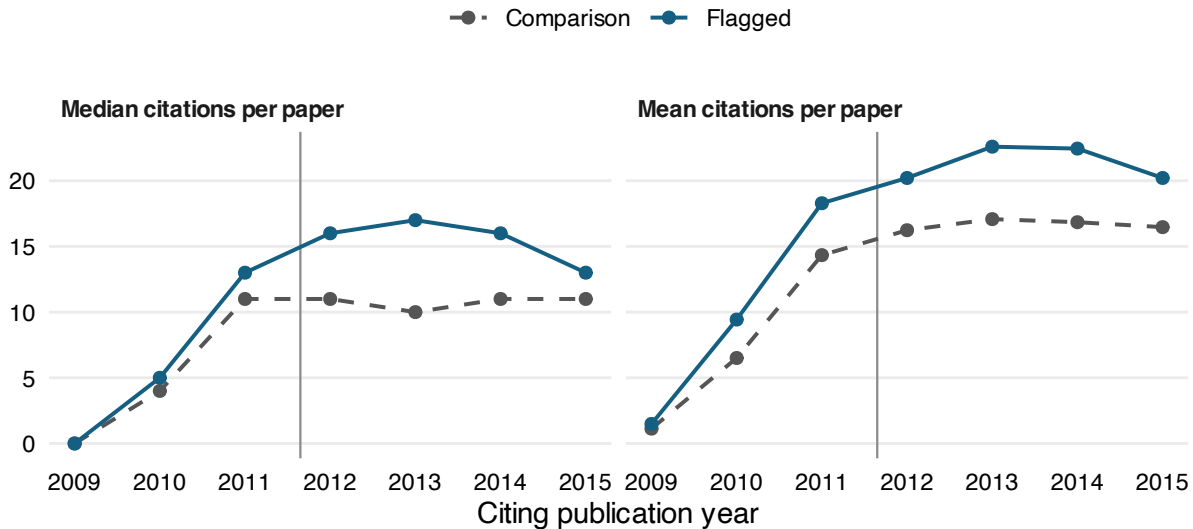
Interpreting that contrast as an effect of publicity requires comparable proportional citation trajectories without the critique. Error status was not randomized, and papers may differ in topic,

prominence, and ordinary citation growth. A general warning could also affect both groups; the comparison concerns their differential response, not the critique’s total influence on science.

### 3 Continued citation and qualification

#### 3.1 Citation trajectories

Continued citation extends beyond a few prominent papers. Flagged papers’ median annual citations rose from 5 in 2010 to between 13 and 17 in each year from 2012 through 2015 (Figure 1). Comparison papers’ medians rose from 4 to 10–11. Mean citations also rose: from 9.4 to a post-period annual average of 21.4 for flagged papers, and from 6.5 to 16.7 for comparison papers. For 91% of flagged papers, the post-period annual average exceeded the baseline.



**Figure 1:** Median and mean annual citations to 76 flagged and 77 comparison papers, including citation-free years. The vertical line marks August 2011, when the critique appeared. The 2009 series includes prepublication zeros for papers published in 2010; the regression baseline is 2010. Lines describe the covered papers rather than a probability sample of all research.

The main regression estimates a 11.5% smaller post/pre citation ratio for flagged papers (95% interval [-31.6, 14.4]%). Thus, widespread continued citation coexists with considerable uncertainty about a relative citation penalty. The medians establish that continued use was typical, while the regression addresses how growth compares with other papers. Neither an observed increase nor an imprecise comparison establishes that publicity had no effect.

Allowing citation trajectories to differ by journal and publication cohort produces estimates from -11.4% to -7.7%. Absolute citation gains instead favor flagged papers by 1.8 citations per paper per year (95% interval [-2.1, 5.7]): flagged papers started from a higher citation level, so absolute and proportional changes answer different questions. The appendix reports the full comparisons, later response windows, and bounds on the magnitude of a relative decline.

### 3.2 Qualifications in citing papers

The archived citation-context sample contains 100 relationships selected after 2011, including 7 from the incomplete 2016 export. All concern citations to flagged papers. Of 95 valid completed ratings, 94 recorded no acknowledgment of concerns and 1 recorded acknowledgment. Unqualified citation was therefore common in the observed sample. The protocol records any concern about the cited paper, not only the specific interaction error.

A citation without criticism need not endorse the faulty inference. It may use another finding, a method, or background information. Nor do the codes reveal whether the authors recognized the mistake. Because the coding sample covers only the post-publicity period, it establishes the rarity of recorded qualifications after publicity, not how their frequency changed. Appendix A reports missing ratings and uncertainty.

## 4 Additional evidence from an instrumental-variables audit

A second methodological critique permits a comparison across fields and statistical problems. Lal et al. (2024) reanalyze instrumental-variable designs in 67 political-science papers. We recover citation histories for that entire assessed-paper cohort. These classifications concern instrument strength and sensitivity of inference; they do not establish that the original substantive conclusions are false. The comparison therefore extends the citation design to adverse diagnostics while keeping that distinction from demonstrated errors explicit.

We examine two main classifications. The first flags an effective first-stage  $F$  below 10, a screening threshold rather than a universal criterion for weak instruments. The second flags a reported estimate that is significant at  $p < .05$  under the archived analytic test but not under the archived Anderson–Rubin or applicable  $tF$  procedure. This second comparison is restricted to papers with at least one analytically significant estimate. A paper satisfies a diagnostic if any of its assessed designs does. Table 1 also reports their union and the narrower Anderson–Rubin comparison; these overlapping definitions are alternative analyses of one audit.

The audit appeared online in May 2024, but an earlier manuscript is dated July 2021. We compare 2023 with 2025 and omit 2024. This contrast concerns citation changes around formal publication, not the effect of first exposure to previously unknown criticism. We count OpenAlex articles and reviews, deduplicate within original-paper citation histories, retain observed zero years, and fit the same article/year fixed-effects Poisson specification with article-clustered uncertainty. All included papers were published before the baseline year. Appendix C describes counting and diagnostic checks.

Papers flagged by the audit continued to receive citations. Under the two main definitions, their mean annual citations rose from 7.0 to 7.2 and from 7.1 to 10.3 between 2023 and 2025; their medians changed little. The comparisons with other assessed papers leave uncertain how much, if at all, the audit reduced citation growth. Both main intervals span reductions and increases, and the estimates change substantially when measured from an earlier year (Appendix C). The opposite signs in Table 1 do not establish that researchers responded differently to different statistical problems.

**Table 1:** Citation changes around the 2024 instrumental-variables audit

| Diagnostic              | Flagged | Comparison | Change (%) | 95% interval   |
|-------------------------|---------|------------|------------|----------------|
| Effective F below 10    | 8       | 59         | -14.5      | [-36.0, 14.2]  |
| Inferential sensitivity | 11      | 44         | 23.9       | [-7.1, 65.3]   |
| Either diagnostic       | 17      | 50         | 11.8       | [-14.6, 46.3]  |
| AR-only sensitivity     | 3       | 52         | -44.0      | [-62.6, -16.0] |

Change is  $100(e^{\hat{\beta}} - 1)$ , comparing flagged and comparison papers’ 2025/2023 citation ratios. Intervals use article-clustered covariance and  $t_{G-1}$  critical values. The inferential-sensitivity and AR-only rows restrict the comparison population to papers with an analytically significant estimate. The rows overlap and are not independent studies. AR-only inference is especially fragile because only 3 papers meet that diagnostic.

The timing also differs from the neuroscience study: the IV audit names the papers it assesses and circulated before formal publication. The historical neuroscience exports and the IV citation histories also come from different databases. Full-cohort comparisons with OpenAlex and OpenCitations yield similar neuroscience relative-growth point estimates (Appendix C).

## 5 An audit of pseudoreplication in animal studies

Lazic et al. (2018) examined 200 animal studies in which interventions were applied to parents and outcomes measured in offspring. They classified 91 as using pseudoreplication, 45 as correctly analyzed, and 64 as too unclear to judge. Pseudoreplication treats dependent observations, such as offspring from the same treated litter, as independent experimental replications. These are the auditors’ assessments of the studied designs; the deposited data do not supply corrected effect sizes.

We recover the entire identified sample and complete OpenCitations histories for all 200 papers. The preprint and paper-level dataset became public in September 2017; journal publication followed in April 2018. Our primary comparison uses 2016 and 2019, excluding the unclear assessments and one flagged paper publicly criticized for the same problem in 2013. This leaves 90 flagged and 45 comparison papers.

Mean annual citations changed from 3.46 to 3.24 among flagged papers and from 4.96 to 4.82 among comparison papers. The corresponding medians were 3 and 2, versus 4 and 4. The difference in mean changes is -0.08 citations per paper (95% interval [-1.22, 1.07]). Citation levels changed little on average in both groups; this comparison supplies no clear evidence of an additional decline among flagged papers. Appendix D reports trajectories, proportional changes and sensitivity checks. As in the other audits, attributing the difference to publicity requires comparable citation paths in its absence.

## 6 Individually publicized errors

The Institute for Replication extension currently supports matched comparisons for only 3 affected papers. This is a pilot, too small and selected to establish how citations generally respond to publicized errors. The source inventory is much larger, but most records still lack a verified conse-

quential error, an established first-disclosure date, sufficient citation history, or complete controls. Appendix F retains the case comparisons and synthetic-control checks.

## 7 Discussion

The affected papers continued to circulate after their statistical mistake was publicized. This pattern appears in median as well as mean citations and is accompanied by very little recorded qualification in the context sample. The evidence is inconsistent with wholesale abandonment of these papers. It is less informative about how many citations publicity prevented.

Citation growth already differed before the critique. Among papers published in 2009, flagged papers’ increase from 2009 to 2010 exceeded that of comparison papers by 4.3 citations (95% interval [0.1, 8.6]). That short publication-year baseline cannot establish the later counterfactual trajectory. Awareness is also unobserved, and some later citations may concern findings unaffected by the mistake. These limits leave room for a citation penalty even as the affected papers remain widely used.

The practical question is how methodological criticism reaches the point of use. Readers can recognize an error from the original analysis, but doing so requires connecting a general warning to a particular paper. Our results document continued citation with little recorded acknowledgment; they do not identify which step in that process accounts for it.

## References

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## A Data construction and citation-context records

The supplied classification spreadsheet identifies the articles by journal, publication year, page, and study characteristics. The identified Excel and CSV versions agree on the paper-level YES/NO codes. The citation import reads 16,437 database records across the covered sheets, repairs 1 malformed record, and marks 15 duplicate DOI records within cited papers. It removes 3 citation relationships identified as false in the historical context coding. Full source hashes and record-level indicators are generated by the replication scripts.

The unreliable histories are source IDs 23 and 25. Matching the intended papers’ journal, year, and supplied page locates the target papers themselves in their supposed citing-article exports. Those histories also contain 54 records before the target publication year. This combination is consistent with exporting literature-search results instead of the intended citation results. The saved workbook contains no query log with which to reconstruct the precise operation. Deleting only the early records would not establish the validity or completeness of the remaining links. The principal estimates exclude both histories, while the appendix reports estimates retaining them. The other sheets contain neither a target-self match nor a record before the target year under these checks.

Missing sheets, source IDs 95 and 124, remain missing. In covered histories the completed 2010–2015 panel includes 29 paper-years with zero recorded citations. The scripts check that every paper has each required year and that joining the classification does not multiply or discard citation records.

**Table 2:** Disposition of the historical citation-context sample

| Recorded status     | Citation relationships |
|---------------------|------------------------|
| Article unavailable | 1                      |
| Concern recorded    | 1                      |
| False citation link | 3                      |
| No concern recorded | 94                     |
| Uncoded             | 1                      |

The sample consists of citation relationships from 2012 through the partial 2016 export. Every relationship matches a record in the source workbooks. “No concern recorded” and “Concern recorded” reproduce completed historical codes for valid citation relationships. The other categories preserve missingness and known false links.

The original coding protocol instructed the rater to assess whether a citing paper noted concerns about the cited paper, using the citing text and original article. The distributed file contains the rating and selected contextual text, not the full highlighted PDFs. We retain those ratings rather than claiming to have independently recoded the articles. The available materials contain one set of item-level ratings, so intercoder agreement cannot be independently checked.

Among valid completed ratings, the acknowledgment share is 1.1% (95% Wilson interval [0.2, 5.7]%). Assigning both unresolved, non-false records to either category gives a share between 1.0% and 3.1% among all non-false sampled relationships. These are missing-rating bounds, not sampling intervals. The Wilson interval is a binomial reference interval for sampled citation relationships; repeated source papers, selection, and coding errors are not represented by it.

## B Models, sensitivity, and uncertainty

### B.1 Proportional changes

For paper  $i$  in year  $t$ , the main model is

$$E[C_{it} \mid i, t, F_i] = \exp\{\alpha_i + \lambda_t + \beta F_i P_t\},$$

where  $F_i$  denotes a flagged paper and  $P_t$  denotes 2012–2015. Article effects absorb stable differences in citation levels and year effects absorb common calendar-year changes. Poisson pseudo-maximum likelihood retains zero counts and does not require equality of conditional mean and variance for the robust inference used here (Santos Silva and Tenreiro, 2006). The reported percentage is  $100(e^\beta - 1)$ , a relative post/pre citation change rather than a percentage change in the number of additional citations gained. In this balanced baseline specification,  $e^{\hat{\beta}}$  equals the ratio of the groups’ post/pre mean citation ratios.

The covariance clusters score contributions by paper. We apply the  $G/(G-1)$  and  $(N-1)/(N-K)$  corrections, counting fixed effects not nested within article clusters in  $K$ , and use  $t_{G-1}$  critical values. Table 3 reports both two-sided 95% intervals and one-sided 95% lower bounds, transformed from the log scale. These are conditional model-based comparisons, not randomized-treatment intervals. No papers are excluded from the main fit for zero outcomes.

**Table 3:** Proportional citation comparisons

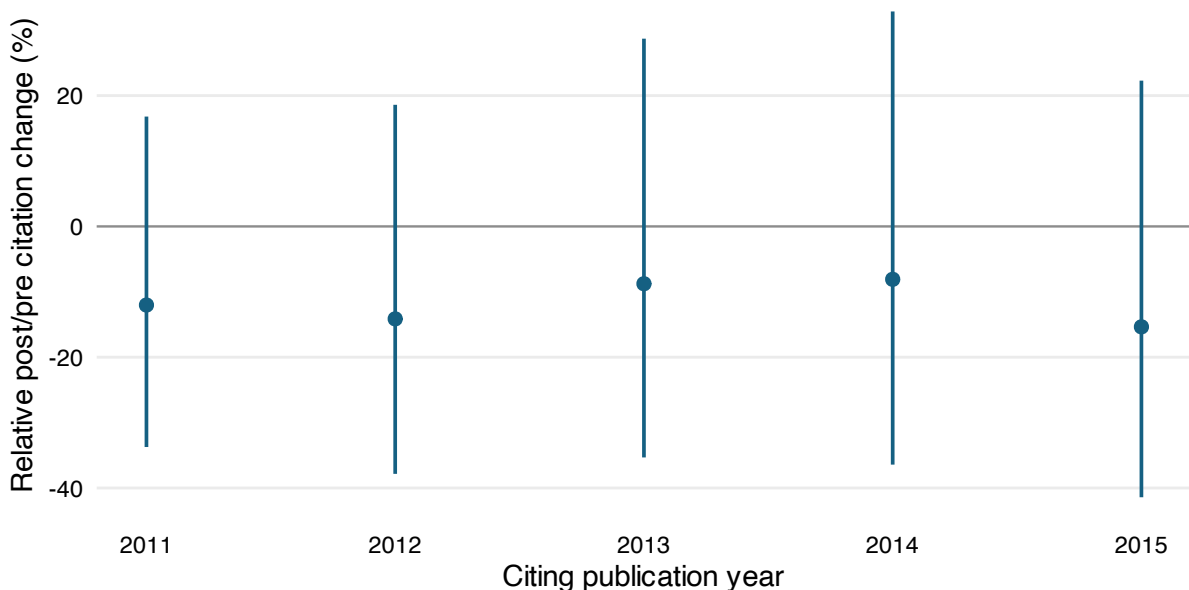
| Specification                        | Flagged / other | Change (%) | 95% interval  | Lower bound |
|--------------------------------------|-----------------|------------|---------------|-------------|
| Article and year effects             | 76 / 77         | -11.5      | [-31.6, 14.4] | -28.7       |
| Journal-by-year effects              | 76 / 77         | -11.4      | [-31.4, 14.5] | -28.5       |
| Cohort-by-year effects               | 76 / 77         | -7.7       | [-24.4, 12.7] | -22.0       |
| Journal/year and cohort/year effects | 76 / 77         | -8.2       | [-24.8, 12.1] | -22.3       |
| Retain questionable histories        | 78 / 77         | -12.2      | [-31.9, 13.2] | -29.1       |
| Retain duplicates                    | 76 / 77         | -11.6      | [-31.6, 14.3] | -28.7       |
| Retain false links                   | 76 / 77         | -11.5      | [-31.6, 14.4] | -28.7       |
| 2009 publication cohort              | 44 / 47         | -7.2       | [-25.0, 14.9] | -22.3       |
| 2010 publication cohort              | 32 / 30         | -10.4      | [-49.8, 60.2] | -44.8       |
| Post period: 2013–2015               | 76 / 77         | -10.7      | [-31.5, 16.4] | -28.5       |
| Post period: 2014–2015               | 76 / 77         | -11.7      | [-32.7, 16.0] | -29.7       |
| Potentially serious errors           | 43 / 77         | 12.8       | [-18.3, 55.6] | -13.9       |

Change and bounds are percentages,  $100(e^\beta - 1)$ . The last column is the one-sided 95% lower bound. Every model includes article effects. The first row includes year effects; the next three replace common year effects with journal-by-year effects, publication-cohort-by-year effects, or both. Other rows use the first row’s specification with the stated inclusion or post-period change. The baseline is always 2010. The potentially-serious subgroup retains the same comparison papers and excludes other flagged papers. Standard errors cluster by article.

The main lower bound excludes proportional reductions greater than 28.7% at the one-sided 95% confidence level under this model. This leaves substantial reductions compatible with the data. The journal/cohort-adjusted lower bounds permit reductions of 22.0–28.5%; these bounds do not resolve uncertainty about the counterfactual trend. A stratified bootstrap resampling whole papers within each group gives a two-sided percentile interval of [-31.2, 14.5]% and a one-sided bound

permitting a reduction of 28.3%. It uses 9,999 draws with seed 31415; no draws are discarded. The baseline ratio identity allows each bootstrap estimate to be calculated directly from the resampled paper-level before and after means.

Year-specific comparisons use 2010 as the baseline (Figure 2). A paper with zero citations in both the baseline and the comparison year has no finite Poisson article intercept and contributes no information to the slope; such papers are excluded from that fit and listed in the generated results. The 2011 contrast describes the transition year. The available history is too short to justify fitting article-specific trends from predominantly post-publicity observations.



**Figure 2:** Year-specific proportional citation changes for flagged relative to comparison papers, using 2010 as the baseline. Points are  $100(e^{\beta} - 1)$  from Poisson article/year fixed-effects models; bars are simultaneous 95% Bonferroni intervals across the five contrasts, with article-clustered uncertainty. Papers with zero counts in both compared years contribute no identifying information and are omitted from the corresponding fit. The 2011 contrast is transitional and is excluded from the main post period.

## B.2 Absolute changes

For an absolute comparison, each paper’s change is its mean annual citations in 2012–2015 minus its 2010 citations. Comparing the two groups’ mean changes gives 1.8 (95% Welch interval [-2.1, 5.7]), allowing unequal variances of paper-level changes. OLS with article and year fixed effects on the same balanced panel reproduces this point estimate exactly; it is another implementation of the same comparison, not independent corroboration. The generated results also report its article-clustered interval. Journal and publication-year adjustment in the paper-level change regression uses HC3 uncertainty.

Removing each source paper in turn gives absolute point estimates from 1.3 to 2.5. A whole-paper stratified bootstrap gives a percentile interval of [-2.0, 5.6]. Medians describe typical citation

**Table 4:** Absolute citation changes under alternative comparisons

| Comparison                                     | Flagged / other | Difference | 95% interval |
|------------------------------------------------|-----------------|------------|--------------|
| Main comparison                                | 76 / 77         | 1.8        | [-2.1, 5.7]  |
| Journal and publication-year adjustment        | 76 / 77         | 0.8        | [-2.8, 4.4]  |
| Retain both questionable citation histories    | 78 / 77         | 1.6        | [-2.3, 5.5]  |
| Retain duplicate database records              | 76 / 77         | 1.8        | [-2.2, 5.7]  |
| Retain known false citation links              | 76 / 77         | 1.8        | [-2.1, 5.7]  |
| Papers published in 2009                       | 44 / 47         | 2.5        | [-2.3, 7.4]  |
| Papers published in 2010                       | 32 / 30         | 0.3        | [-6.4, 7.0]  |
| Post period: 2013-2015                         | 76 / 77         | 2.0        | [-2.2, 6.2]  |
| Post period: 2014-2015                         | 76 / 77         | 1.7        | [-2.5, 6.0]  |
| Potentially serious errors only                | 43 / 77         | 4.3        | [-0.1, 8.7]  |
| Pre-critique change: 2009 cohort, 2009 to 2010 | 44 / 47         | 4.3        | [0.1, 8.6]   |

Differences are flagged minus comparison, in citations per paper per year. Except for the final row, the baseline is 2010; the post period is 2012–2015 unless otherwise stated. Intervals are pointwise 95% Welch intervals, except the journal/year-adjusted estimate, which uses OLS with HC3 standard errors and a residual-degrees-of-freedom  $t$  critical value. The final row compares 2010 with 2009 among papers published in 2009; it is a diagnostic, not a publicity-effect estimate.

levels; neither medians nor these sensitivity checks make error status or publicity exposure random. All analytical choices are retrospective.

Of the distinct citing documents in the main papers’ 2010–2015 records, 13.7% cite more than one study paper. Shared citing documents and related topics can induce dependence across papers. Article clustering and the whole-paper bootstrap do not model those connections. The source review is also a selected set of journals and years, so these comparisons are not probability-sample estimates for all scientific publishing.

## C Additional-cohort construction and comparability

The IV replication archive contains multiple designs for some papers. We retain the design records but assign each paper one citation history and equal weight within its diagnostic group. The repository reproduces the point estimates for the 10 designs selected in the separate citation-context pilot. For the Alt design, the instrument is an eight-level factor with seven excluded indicators. Recomputing the joint Anderson–Rubin test with that encoding corrects the archived calculation; its loss-of-significance classification remains the same. The single-instrument  $tF$  calculation is inapplicable and does not enter that design’s sensitivity classification. These checks do not constitute independent replication of every diagnostic in the full audit.

The relative citation-growth estimates are -14.5% for the weak-instrument screen and 23.9% for inferential sensitivity. On a fixed older cohort, moving the baseline from 2023 to 2022 changes the weak-instrument estimate from -11.5% to -0.4%, and the sensitivity estimate from 15.7% to -17.0%. Holding the cohort fixed shows that changing the starting year alone can substantially change the comparison.

Document types materially affect the IV comparison. The original broad outcome included preprints, proceedings and book chapters; some books contributed links from numerous chapters

and front matter to the same original paper. After observing this problem and the alternative estimates, we selected articles and reviews as the primary outcome. For the weak-instrument screen, the broader count gives 39.3%, compared with -14.5% for articles and reviews. This was an outcome-aware measurement revision. The repository retains the broader results and a partial sensitivity that collapses identified Oxford book DOI families; neither procedure resolves all cross-version duplication.

The source-comparison protocol preserves the original neuroscience analysis cohort and compares identical papers and years before considering a pooled result. Exact DOI matches identify shared citing relationships, with each database’s publication year retained. Historical exports lack the detailed document-type field needed to impose an identical article/review filter. Missing histories stay missing; a relationship absent from one database is not automatically a false citation. Source discrepancies therefore measure differences between the recorded citation histories rather than bias relative to an assumed ground truth.

The complete paired comparison contains 76 flagged and 77 comparison papers. Replacing the historical counts with OpenAlex articles and reviews gives a relative-growth contrast of -10.9% (95% interval [-29.0, 11.9]%), compared with -11.5% in the historical exports. The paired, paper-level bootstrap estimates a source-induced change in the growth ratio of 0.8% (95% interval [-8.9, 12.4]%). The broader OpenAlex document-type count gives a contrast of -10.7%. These are comparisons of recorded citation histories on fixed papers and years; neither database is assumed to be ground truth.

Among 3223 shared citation relationships with conflicting years, OpenAlex dates 3219 earlier. Publisher-deposited online and print dates explain 151 differences in the reviewed links; the remaining dates have not all been independently reviewed. Moving shared eligible citations between years contributes 242 citations to the 431-citation difference in the 2010 totals. Coverage and publication dating can therefore both change measured growth.

For the 153 paired papers checked against OpenCitations, that index also records 2141 of the 2637 OpenAlex-only links eligible for the primary article/review count. Across all document types in the comparison frame, it records 3451 of 4719 OpenAlex-only links. This corroborates many additional records, but shared upstream sources mean agreement across indexes is not independent verification of the citing bibliography. Undated OpenCitations links remain undated, and this check does not replace either source’s annual counts.

A separate OpenCitations comparison covers all 153 historical papers. Its estimated relative-growth contrast is -11.9%, compared with -11.5% in the historical exports. OpenCitations records more citations in both periods, leaving this aggregate comparison almost unchanged. The outcome counts dated citing works across document types; 310 undated relationships remain unassigned. A paired, paper-level bootstrap gives a source-induced change in the estimated growth ratio of -0.4% (95% interval [-8.0, 9.2]%). These intervals condition on the observed cohort and database records. Similar aggregate estimates do not verify individual links or establish either source as ground truth.

For comparison across audits, the repository also reports the contrast between the calendar year before and the calendar year after the warning year, excluding the warning year itself. This uses 2010 and 2012 for neuroscience and 2023 and 2025 for the IV audit. The neuroscience papers

published in 2010 have a partial publication-year baseline, so a separate 2009-publication cohort has a complete baseline year. The synthesis below uses these common windows; alternative IV diagnostics are not counted as independent replications.

## D Pseudoreplication audit: construction and sensitivity

The source sampled eligible English-language prenatal-exposure experiments in multiparous mammals from 500 PubMed search results. Our population is the 200 deposited assessments, not animal research generally. Exact PMID matching identifies all 200 papers; 198 also have a DOI. We retain the 2 papers without DOI and query their citations by PMID. The first preprint appeared on September 2, 2017, and the identified dataset was released on September 6. The repository documents the release dates and source provenance.

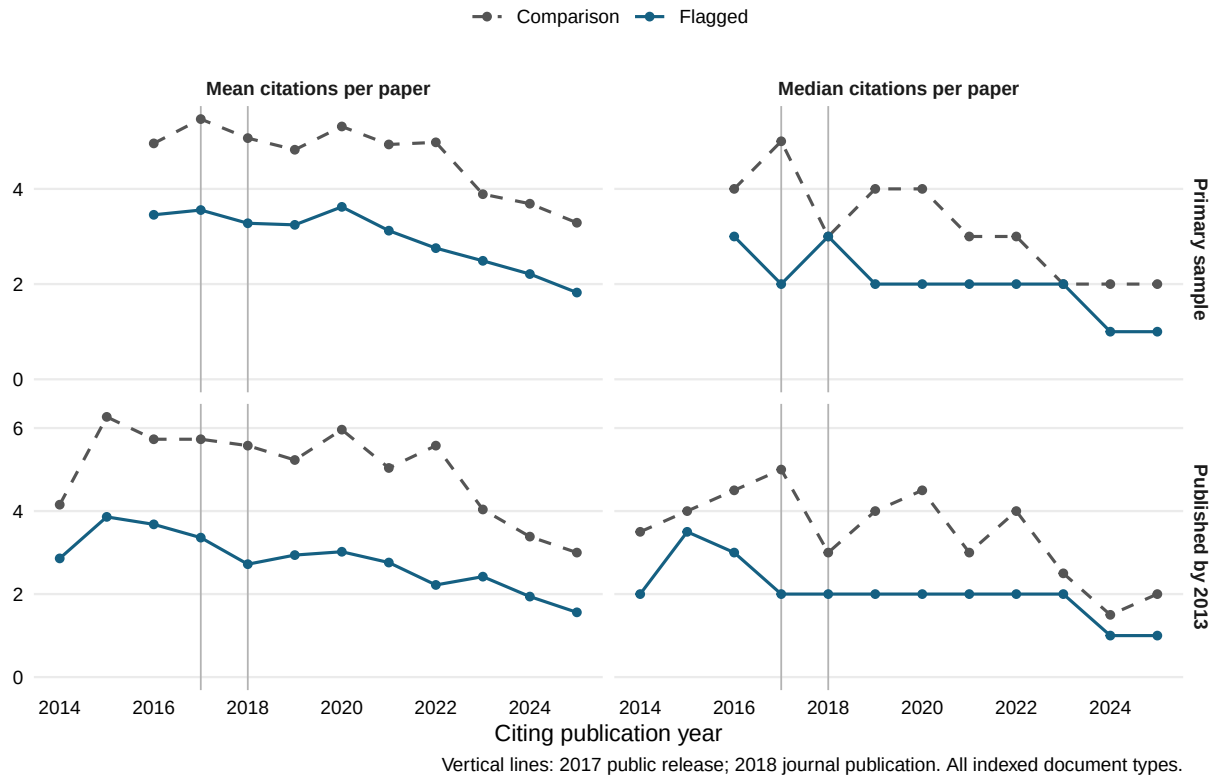
OpenCitations supplies 8818 relationships. Citing records sharing a DOI or OpenCitations identity count once per target; 70 relationships have no assignable year. Counts include all indexed document types, including distinct book chapters. A check of 18 chapter–target pairs in three prominent book families verifies separate bibliographies containing the target references. We retain those chapters. A completed API history permits observed zero years; acquisition failure would not. The repository records identities, source checksums, citation links and notice adjudications.

One flagged paper received a public methodological criticism on May 17, 2013, with a response disputing the criticism. We exclude it from the first-known-publicity comparison and retain all 136 classified papers in a sensitivity. This eligibility change preceded estimation. Earlier errata for other papers include name and affiliation corrections; two notices remain incompletely reviewed. Excluding all papers with linked prior errata is an additional sensitivity, not a claim that every notice concerned the statistical problem.

The main absolute contrast is the difference in equal-paper mean changes, with a Welch interval. Article/year fixed-effects Poisson estimation gives a proportional contrast of -3.5% (95% interval [-25.4, 24.7]%). The 10 papers with zero citations in both selected years drop from that fit but remain in the absolute contrast and descriptive summaries. The full-cohort and prior-erratum exclusions, a 2019–2021 post-period average, and a fixed cohort published by 2013 yield similarly inconclusive comparisons. Split-unit designs are more common among flagged papers; standardizing within those design strata gives a difference of -0.57 citations (bootstrap interval [-1.61, 0.47]). The older cohort also had differing mean citation gains before publicity, weakening a causal interpretation. Full estimates and whole-paper bootstrap intervals are generated in the repository.

## E Provisional synthesis across audits

We combine one proportional contrast from each of the three audits: Nieuwenhuis, Lal, and Lazic. We weight each log relative-growth contrast by the inverse of its estimated variance:  $w_j = s_j^{-2} / \sum_k s_k^{-2}$  and  $\bar{\beta} = \sum_j w_j \hat{\beta}_j$ . The reported quantity is  $100(e^{\bar{\beta}} - 1)$ . It summarizes the included audit effects with greater weight on more precise estimates. We use 2010 versus 2012 for Nieuwenhuis, 2023 versus 2025 for Lal, and 2016 versus 2018 for Lazic. The last contrast is the



**Figure 3:** Mean and median citations in the animal-study audit. The primary sample contains 90 flagged and 45 comparison papers. The older fixed cohort contains 50 and 26, respectively. Vertical lines mark the 2017 public release and 2018 journal publication. Earlier-warning papers and unclear classifications are excluded. Counts include all indexed document types; citation-free covered years remain zero.

year after first public release but the year of journal publication; the repository also uses the main Lazic follow-up in 2019. Earlier circulation of the Lal critique means that its formal publication is not a first-disclosure date.

**Table 5:** Precision-weighted synthesis across the assembled audits

| IV definition           | Change (%) | 95% interval  |
|-------------------------|------------|---------------|
| Effective F below 10    | -12.1      | [-24.5, 2.4]  |
| Inferential sensitivity | -2.1       | [-15.9, 14.0] |

Entries are percentage relative-growth contrasts with 95% intervals. Each row uses one IV diagnostic; overlapping definitions are alternatives, not independent audits. Inverse-variance weights and normal intervals condition on the included audits and estimated component variances. The I4R pilot is excluded from pooling.

The precision-weighted estimates are -12.1% under the instrument-strength screen and -2.1% under inferential sensitivity. The intervals in Table 5 quantify the remaining uncertainty about magnitude. The source comparisons and alternative follow-up year give similar estimates. A causal interpretation requires that flagged and comparison papers would otherwise have followed comparable proportional citation trajectories within each audit.

The main fixed-effect synthesis has standard error  $1/\sqrt{\sum_j s_j^{-2}}$  and uses normal critical values. It estimates a precision-weighted mean for the included audits without requiring identical effects. The repository reports the audit weights, leave-one-audit-out estimates, equal-audit weighting, and a random-effects sensitivity. The latter estimates between-audit variance by restricted maximum likelihood and uses modified Knapp–Hartung intervals with two degrees of freedom; its standard-error adjustment cannot shrink uncertainty below the unadjusted value. Three audits provide little information about the distribution of effects across critiques. No DOI-bearing Lazic paper overlaps the other original/control papers; one included Lazic paper lacks a DOI.

The I4R pilot contains only 3 selected disclosures and is excluded from every pooled estimate. Its case histories, matched comparisons and synthetic-control checks remain separate. Expansion focuses on complete methodological audits and replication projects with identified original papers and both adverse and favorable assessments.

## F Matched cases with individually publicized errors

The I4R registry separates catalog listings, article-specific assessments and verified error disclosures. Unavailable reports and unnamed reviewer teams remain unresolved records; overview summaries alone do not establish material errors. Verified cases link the original article to the error, affected claim, source passage, replies and earliest substantiated public disclosure. Material corrections can strengthen or weaken a finding; these are not necessarily refuted papers. A later I4R listing does not reset the disclosure clock. The repository’s claim assessments and disclosure ledger provide the case-level evidence.

The secondary analysis uses cached OpenAlex work records with verified identities, source hashes, retrieval dates and supported annual coverage. Missing histories remain missing. Controls exclude known assessed papers, retracted papers and verified nonresearch items. Matching waits

for a complete eligible journal pool and uses up to three controls, selected by pre-disclosure citation levels, growth and text distance with fixed calipers. The matching rules and measurement choice were recorded before these contrasts were estimated; this remains a retrospective analysis. Post-disclosure outcomes do not enter matching.

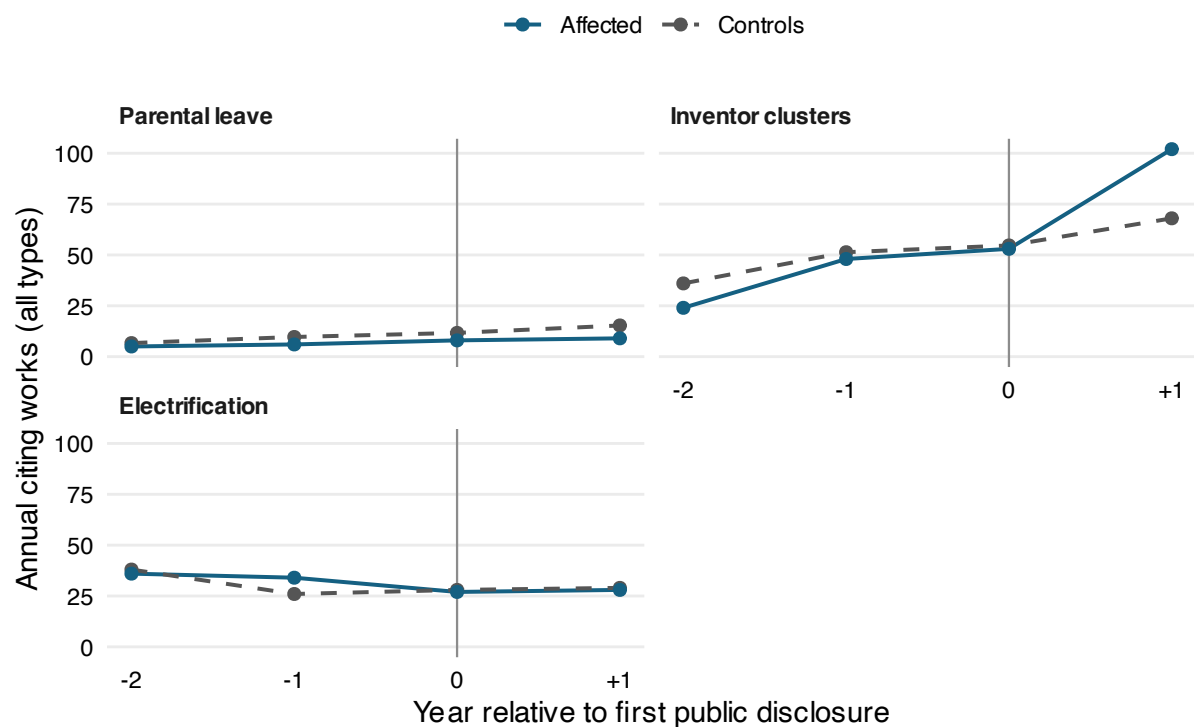
**Table 6:** Citation changes after individually publicized errors

| Paper             | Years     | Affected             | Controls                | Contrast |
|-------------------|-----------|----------------------|-------------------------|----------|
| Parental leave    | 2020–2022 | 6 $\rightarrow$ 9    | 9.7 $\rightarrow$ 15.3  | -2.7     |
| Inventor clusters | 2023–2025 | 48 $\rightarrow$ 102 | 51.3 $\rightarrow$ 68.0 | +37.3    |
| Electrification   | 2020–2022 | 34 $\rightarrow$ 28  | 26.0 $\rightarrow$ 29.0 | -9.0     |

Annual citing works of all document types. The years column gives the calendar year before and after first documented disclosure; the disclosure year is excluded. Controls are equally weighted within each matched set. The final column subtracts the control change from the affected paper’s change. These are individual observed contrasts, not independently identified causal effects.

Mean citations rose from 29.3 to 46.3 among affected papers and from 29 to 37.4 among controls. The equal-paper average difference in changes is 8.6 citations (exploratory 95% interval [-56.9, 74]). First differences with matched-set effects reproduce this average. The covariance clusters by original article and disclosure event, retaining original identities when controls recur; the critical value uses the smaller cluster count minus one as degrees of freedom. Inference conditions on selected matches and is fragile with only 3 disclosures.

The preceding year shows a matched growth difference of +5.6 citations, despite matching on pre-period outcomes. Figure 4 shows the differing trajectories. Leaving out the inventor-clusters case changes the average post-disclosure contrast to -5.8. The repository reports all leave-one-disclosure-out contrasts, alternative baselines and longer horizons on common cohorts. These diagnostics prevent treating the positive average across these cases as evidence that publicity increases citations. They also show why continued citation alone cannot measure the citations that publicity prevented.



**Figure 4:** Citation paths for the supported I4R cases. Solid lines show each affected paper; dashed lines show its equally weighted matched controls. Counts include citing works of all types. Year zero marks the calendar year of first documented public disclosure, not its exact date. All panels use the same scale. These descriptive counts have no sampling intervals and do not measure endorsement of the erroneous claim.

## F.1 Synthetic-control sensitivity

We also fit nonnegative donor weights summing to one to match each affected paper’s pre-disclosure citation path. Donors retain the journal, publication-age, identity and exposure restrictions, but the citation caliper and text ranking are removed. We require at least three full pre-years, first withholding the last year to check prediction and then refitting on all pre-years. This supports 2 cases; inventor clusters has only two full pre-years. A fixed small ridge penalty resolves the many possible donor combinations. All choices are retrospective.

**Table 7:** Synthetic-control fit and prediction checks

| Paper           | Pre-fit RMSE | Held-out: actual / predicted | Post: actual / synthetic | Gap  |
|-----------------|--------------|------------------------------|--------------------------|------|
| Parental leave  | 0.0          | 6.0 / 8.4                    | 9.0 / 9.6                | -0.6 |
| Electrification | 0.0          | 34.0 / 40.5                  | 28.0 / 35.2              | -7.2 |

Annual all-type citations. Pre-fit RMSE describes the final fit using all pre-years; the held-out prediction uses only earlier years. Post is the calendar year after disclosure. The gap is actual minus synthetic citations. No sampling or randomization interval is implied. Full donor weights, paths and sensitivity estimates are in the repository.

The synthetic comparisons are mixed and do not resolve the counterfactual problem. Parental leave is close to its synthetic comparison and electrification is below it. The held-out pre-year predictions exceed actual citations in both cases. Close in-sample fit therefore does not establish a reliable untreated trajectory, and these gaps are not pooled into a causal estimate.